

diff - diff

Callaway-Sant'Anna. Cengiz-style stacking.

BJS-style imputation. The classic 2x2.

**One regression
to rule them all.**

LP-DiD: Local Projections DiD.

Now in diff-diff.

Dube, Girardi, Jordà & Taylor (2025).

Journal of Applied Econometrics 40(5)

The staggered revolution left you with a shelf.

Classic 2x2 DiD

Where you started.

One clean comparison.

Callaway-Sant'Anna

Group-time ATTs.

Its own aggregation layer.

Cengiz-Style Stacking

Cengiz et al. (2019).

Its own dataset duplication.

BJS-Style Imputation

Borusyak-Jaravel-Spiess.

Its own two-step machinery.

Three fixes for TWFE's negative weights - plus the textbook case they outgrew.

Same causal question. Four mental models.

Same question.

Four different machines.

Staggered State Policy

Minimum wage rolls out state by state. CS? Stacked? Which aggregation?

Phased Product Rollout

The feature ships to markets in waves. Event study? Which estimator?

Programs That Switch Off

The retention treatment ends. Absorbing-only estimators don't apply.

Survey-Weighted Panels

CPS-style pweights, strata, PSUs. Which estimator takes a design?

LP-DiD.

It's just a regression.

A local projection per horizon: long difference,
time fixed effects, clean controls. Plain OLS.

One long difference per horizon.

OLS on clean controls. You can read every comparison it makes.

Weights provably non-negative.

Clean-control baseline result (Eqs. 9-10). With covariates, prefer RA.

The estimand is a dial.

Variance-weighted for precision, or equal-weighted - your call.

Local projections (Jordà 2005), pointed at DiD.

Clean controls, horizon by horizon.



Each horizon is one regression you could run by hand.

In the absorbing (default) path, already-treated units never serve as controls.

The zoo was one regression all along.

$$y_{i,t+h} - y_{i,t-1} = \beta_h \Delta D_{it} + \delta_t^h + e_{it}^h$$

$$E(\beta_h) = \sum_{g \neq 0} \omega_{g,h} \tau_h^g, \quad \omega_{g,h} \geq 0$$

always non-negative -- no forbidden comparisons



<code>reweight=True</code>	== Callaway-Sant'Anna	<i>exact (Sec. 3.7)</i>
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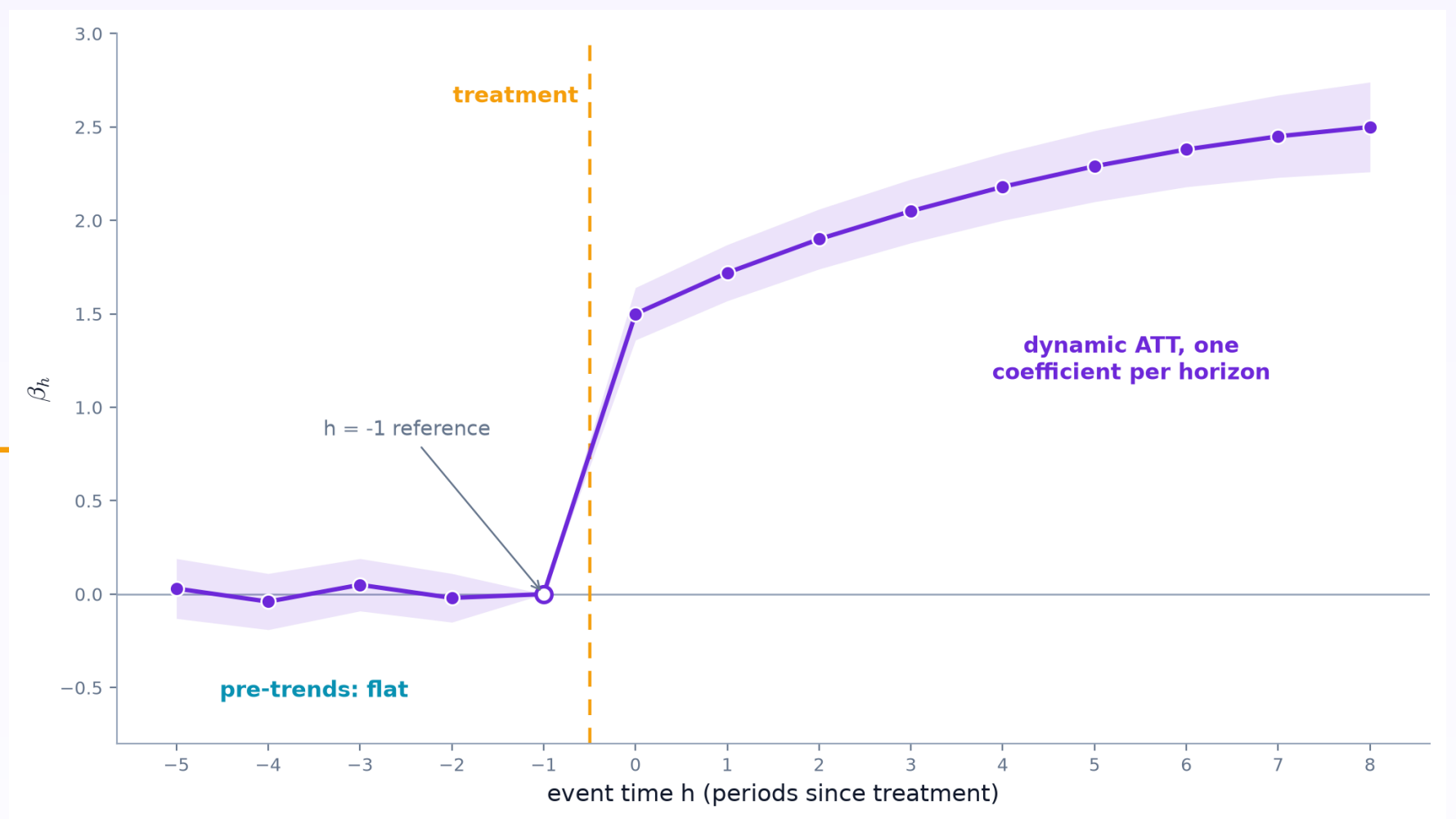
<code>default (variance-weighted)</code>	== Cengiz et al. stacked	<i>exact (Sec. 3.7)</i>
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<code>h=0, single cohort</code>	== classic 2x2 DiD	<i>exact (Sec. 2.2)</i>
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<code>pmd= 'max '</code>	-> BJS imputation	<i>exact single-cohort (fn. 10-11)</i>
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Numerical equivalences proven in the paper - not analogies.

Pre-trends are just negative horizons.



The event study is native - not a post-hoc aggregation.

The Code.

Same sklearn-like API as every diff-diff estimator.

```
from diff_diff import LPDiD

result = LPDiD(
    pre_window=5, post_window=10,
).fit(
    data,
    outcome='revenue', unit='store',
    time='week', treatment='treated',
)

print(result.att)          # pooled post ATT = 2.31
result.event_study         # one beta per horizon

# flip one switch:
#   reweight=True   -> Callaway-Sant'Anna
#   non_absorbing=  -> on/off treatments (entry effects)
#   survey_design=  -> pweights + strata + PSUs (default path)
```

Callaway-Sant'Anna is literally a keyword argument.

Production-ready.

Cluster-Robust SEs

Unit-level by default,
matches Stata `lstdid`

Covariates, Two Ways

RA path (BJS-style) or
direct with guardrails

PMD Baselines

Premean differencing +
pooled pre/post ATTs

Non-Absorbing Treatment

First-entry & effect-
stabilization (Eq. 12/13)

Survey Designs

`pweights`, `strata`, `PSU`, `FPC`;
Binder TSL (default path)

Composition Control

`no_composition` fixes the
post-window sample

Composable where validated - `survey_design` runs the default path only.

Validated.

Against the authors' own tooling - and against our own shelf.

Authors' R recipes (danielegirardi/lpdid)

Event-study and pooled estimands match to $\sim 1e-12$.

Independent fixest reconstruction

Non-absorbing Eq. 12/13, variance-weighted: point + SE to $\sim 1e-13$.

survey::svyglm, end to end

Default-path survey: per-horizon point, SE, and df all pinned.

Equivalences tested, not cited

reweight == our CallawaySantAnna; PMD == our ImputationDiD (1 cohort).

Now in diff-diff.

LP-DiD.

```
$ pip install --upgrade diff-diff
```

github.com/igerber/diff-diff

diff - diff

Difference-in-Differences for Python